

10. AI Large Model Offline Voice Assistant

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1. Offline Voice Configuration
2. Creating a Boot Auto-start Service (Systemd)
 - 2.1 Creating a Startup Script
 - 2.2 Creating the Systemd Service File
 - 2.3 Managing and Debugging the Service

1. Offline Voice Configuration

Before setting up auto-start, we must ensure that the program itself can work independently in an offline state. This needs to be done by modifying configuration files.

1. Locate the configuration file:

In your project code, find and open the configuration file:

```
config/yahboom.yaml
```

2. Modify configuration parameters:

Please check the following parameters in the file and ensure their values match those shown below. If the parameters do not exist, please add them.

```
asr:                                     #Voice node parameters
  ros__parameters:
    VAD_MODE: 2                         #VAD sensitivity
    sample_rate: 16000                  #ASR recording audio sample rate
    frame_duration_ms: 30               #VAD frame size in ms
    use_online_asr: False               #Whether to use online ASR
  recognition (True for online, False for offline)
  mic_serial_port: "/dev/ttyUSB0"      #Microphone serial port alias
  mic_index: 0                         #Microphone index
  language: 'zh'                       #ASR language

model_service:                          #Model server node parameters
  ros__parameters:
    language: 'zh'                     #Large model interface language
    useolinetts: False                 #Whether to use online speech
  synthesis (True for online, False for offline)

  # Large model configuration
  # llm_platform: 'ollama'              # Optional platforms: 'ollama',
  'tongyi', 'spark', 'qianfan', 'openrouter'
  llm_platform: 'ollama'               # Currently selected large model
  platform
```

- `use_online_asr` and `useolinetts` must be set to False.
- `llm_platform` must be set to `ollama`.

With this configuration, everything will be offline.

3. **Save the file** and **rebuild** the project to apply changes:

```
cd ~/yahboom_ws
colcon build
source install/setup.bash
```

After completing this step, the program is already a purely offline voice service.

2. Creating a Boot Auto-start Service (Systemd)

Now, we will create a `systemd` service to make `largemodel_control.launch.py` run automatically at system startup.

2.1 Creating a Startup Script

To enable `systemd` to correctly load the ROS2 environment, the best practice is to create a simple `bash` script to encapsulate our startup command.

1. **Create the script file:**

In the directory (`~/yahboom_ws/src/largemodel/`), create a file named `start_largemodel.sh`.

```
vim ~/yahboom_ws/src/largemodel/start_largemodel.sh
```

2. **Write the script content:**

Copy and paste the following content into the script file.

```
#!/bin/bash

# Source ROS2 Jazzy environment
source /opt/ros/jazzy/setup.bash

# Source Yahboom workspace environment
source /home/sunrise/yahboom_ws/install/setup.bash

# Start largemodel control script
ros2 launch largemodel largemodel_control.launch.py
```

Important: Please ensure to replace `/home/sunrise/` in the script with your own user home directory path.

3. **Save and exit**
4. **Grant execute permission to the script:**

```
chmod +x ~/yahboom_ws/src/largemodel/start_largemodel.sh
```

2.2 Creating the Systemd Service File

This is the most critical step. We will tell the system that we have a new service to manage.

1. **Create the service file:**

You need to use `sudo` privileges to create this file.

```
sudo vim /etc/systemd/system/largemodel.service
```

2. Write the service configuration:

Copy and paste the following content completely into the service file.

```
[Unit]
Description=Robot Service
After=network.target sound.target graphical.target multi-user.target
Wants=network.target sound.target graphical.target multi-user.target

[Service]
Type=simple
User=sunrise
Group=sunrise
Environment=DISPLAY=:0
Environment=XDG_RUNTIME_DIR=/run/user/1000
Environment=PULSE_SERVER=unix:/run/user/1000/pulse/native
SupplementaryGroups=audio video
ExecStartPre=/bin/sleep 10
ExecStart=/home/sunrise/yahboom_ws/src/largemodel/start_largemodel.sh
Restart=on-failure
StandardOutput=journal
StandardError=journal

[Install]
WantedBy=multi-user.target
```

- The path in `ExecStart` must exactly match the startup script path.

3. Save and exit.

2.3 Managing and Debugging the Service

Now that your service is created, we need to let `systemd` load it and set it to auto-start on boot.

1. Reload the `systemd` daemon to read our newly created service file:

```
sudo systemctl daemon-reload
```

2. Set the service to auto-start on boot:

```
sudo systemctl enable largemodel.service
```

3. Start the service immediately:

```
sudo systemctl start largemodel.service
```

4. Check service status:

This is the most important command to verify whether the service is running successfully.

```
sudo systemctl status largemodel.service
```

- If you see `Active: active (running)`, congratulations, the service has started successfully!

- If the status is `failed` or other, please continue to the next step for debugging.

5. View service logs (essential for debugging):

If the service fails to start, you can view all real-time logs generated by the `ros2 launch` command through the following command, which is crucial for locating problems.

```
journalctl -u largemodel.service -f
```

After completing all the above steps, every time the system boots, it will automatically start the purely offline `largemodel` voice service.

To disable the online voice assistant service, run the following commands

```
sudo systemctl disable largemodel.service
```

```
sudo systemctl stop largemodel.service
```